

Supporting Information

Null-Model Choice for Rule-Defined Pattern Counts in Clustered Longitudinal Sequences: A Case Study of Jewish Halachic Menstrual Anticipation Rules

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This document contains the exploratory between-woman pairwise dependence analysis referred to in Sections 3.4, 4.4 and 5.2 of the main article (Section S1), the literature search referred to in Section 1 (Section S2), the evaluation of the zero-count rule *Haflaga Chozer Chalila* referred to in Sections 2.1 and 3.1 (Section S3), the full results of the external replication in the Utah Creighton Model cohort referred to in Sections 3.6 and 4.6 (Section S4), the null distribution functions under the two global nulls (Section S5), and the regularity-stratified and per-woman checks of the within-woman null referred to in Sections 4.2 and 4.6 (Section S6). The pairwise analysis is secondary to the null-model comparison and does not bear on the primary question of whether within-woman ordering produces excess rule-defined events. Notation follows the main article; all computations are produced by `scripts/postprocess.py` and the accompanying notebook in the code repository (<https://github.com/dvirross/VesetVectorTest>).

The SHA-256 checksum of the released filtered data file (`data/FilteredData.csv`; 1,554 cycles, 118 women), for verifying file integrity, is

508ee88efb18bcd29c7ed6f841827377bb8d72a8c7dd38fdeb40cf08691f7811

S1. Exploratory Pairwise Dependence Between Pattern Types

S1.1 Question and expectations

The joint test of the main article treats the five counts as one vector. The pairwise analysis asks a different and secondary question — whether women who produce one pattern type tend to produce another — as a consistency check against the relationships derived from the pattern definitions (main article, Section 2.2). Those relationships lead to two expectations: a positive $\text{Haflaga} \times \text{Week-Dilug}$ association, which is partly an arithmetic consequence of containment

(every Haflaga sequence at $H = 30$ contains a Week-Dilug event) rather than an empirical hypothesis, and a plausible but not entailed positive Haflaga $\times$ Dilug-in-Dilug association from polynomial proximity. No expectation is held for the other eight pairs, so the analysis is a consistency check, not a search.

S1.2 Methods and rationale

Unit of analysis: the woman. — Pattern events are defined within a woman’s sequence and cycles within a woman are not independent, so the woman is the only exchangeable unit for an association test. The analysis works with 118 per-woman summaries, not with 1,554 cycles.

Rates rather than counts. — A woman with 45 cycles has roughly nine times as many opportunities to produce *any* pattern as a woman with 5 cycles, so raw per-woman counts of two patterns are positively correlated through follow-up length alone. Dividing each count by the number of cycles removes the first-order part of this exposure effect. It does not remove it entirely: a rate from 5 cycles is far noisier than one from 45, the minimum run lengths (2, 3 or 4 cycles) mean short sequences have proportionally fewer eligible windows, and in these data the Haflaga rate itself correlates with follow-up (Spearman correlation with n_i : 0.25 for Haflaga, 0.13 for Week, 0.10 for Dilug-in-Dilug, 0.02 for Dilug, 0.00 for Week-Dilug). The permutation null below is therefore one of *marginal* independence between two rates, not of independence conditional on follow-up; Section S1.4 checks how much this matters.

Spearman rather than Pearson correlation. — Per-woman rates are bounded at zero, heavily zero-inflated (most women have no event of a given type) and right-skewed. Pearson’s coefficient would be dominated by the few women with several events; the rank correlation measures monotone association and is insensitive to the magnitude of those few values.

Permutation rather than asymptotic p -values. — With 118 women, many tied zeros and skewed rates, the usual t -approximation for the null distribution of r_s is unreliable. One rate vector is permuted across women 50,000 times (seed 17); this breaks any association between the two patterns while preserving both marginal distributions exactly, giving a two-sided Monte Carlo p -value (proportion of permutations with $|r_s^*| \geq |r_s|$, with the $(b+1)/(B+1)$ convention).

Holm correction over ten pairs. — Five patterns yield ten pairs, of which only two carry a structural expectation. Holm’s step-down procedure controls the family-wise error rate at $\alpha = .05$ over all ten and is valid under any dependence among the tests, which matters because the pairs share patterns.

Binary robustness check. — Because the rate metric depends on the exposure normalisation and on the few women with several events, the analysis is repeated with each woman coded 1/0 for ever having established each pattern. Each pair is then a 2×2 table tested with Fisher’s exact test (conditional on the margins) and with a permutation χ^2 statistic (permuting one indicator across women), again Holm-corrected. The probability of ever observing an event also increases with follow-up, so this check addresses the choice of metric, not the follow-up dependence.

S1.3 Results

Table S1 and Figure S1 give the rate-based Spearman correlations. Haflaga $\times$ Dilug-in-Dilug is positive and survives Holm correction ($r_s = 0.27$, $p_{\text{perm}} = .004$, Holm $p = .038$): women with a higher rate of exact repetitions also tend to have a higher rate of second-order progressions, most of which ($|d| = 1$ in 16 of 27) are near-constant sequences. Haflaga $\times$ Week-Dilug is positive, as containment requires, but does not survive correction ($r_s = 0.23$, $p_{\text{perm}} = .011$, Holm $p = .10$); only 3 of the 26 Haflaga sequences are at $H = 30$, so the guaranteed part of the association is small. The remaining eight pairs are null (all Holm $p = 1$), including Dilug $\times$ Week-Dilug, whose weak negative sign ($r_s = -0.12$) is in the direction implied by their mutual exclusion but is not distinguishable from zero.

The binary (ever/never) version reproduces the ordering and the two smallest p -values (Haflaga $\times$ Dilug-in-Dilug: $\chi^2 = 9.99$, $p_{\text{perm}} = .004$, Holm $p = .042$, Fisher $p = .004$; Haflaga $\times$ Week-Dilug: $\chi^2 = 7.26$, $p_{\text{perm}} = .013$, Holm $p = .12$), so the findings do not depend on the rate normalisation or on the few women with multiple events.

Table S1: Pairwise Spearman correlations between per-woman pattern rates ($N = 118$), sorted by p_{perm} (two-sided, 50,000 permutations); p_{Holm} across the 10 pairs. §: pair with an analytically expected positive association (main article, Section 2.2). Bold: $p_{\text{Holm}} < .05$.

Pair	r_s	p_{perm}	p_{Holm}
Haflaga $\times$ Dilug-in-Dilug§	+0.27	.004	.038
Haflaga $\times$ Week-Dilug§	+0.23	.011	.102
Dilug $\times$ Week-Dilug	−0.12	.206	1.00
Haflaga $\times$ Week	+0.10	.277	1.00
Week $\times$ Dilug-in-Dilug	+0.09	.326	1.00
Dilug $\times$ Week	+0.09	.337	1.00
Week $\times$ Week-Dilug	+0.06	.476	1.00
Week-Dilug $\times$ Dilug-in-Dilug	+0.07	.480	1.00
Haflaga $\times$ Dilug	+0.04	.668	1.00
Dilug $\times$ Dilug-in-Dilug	−0.02	.819	1.00

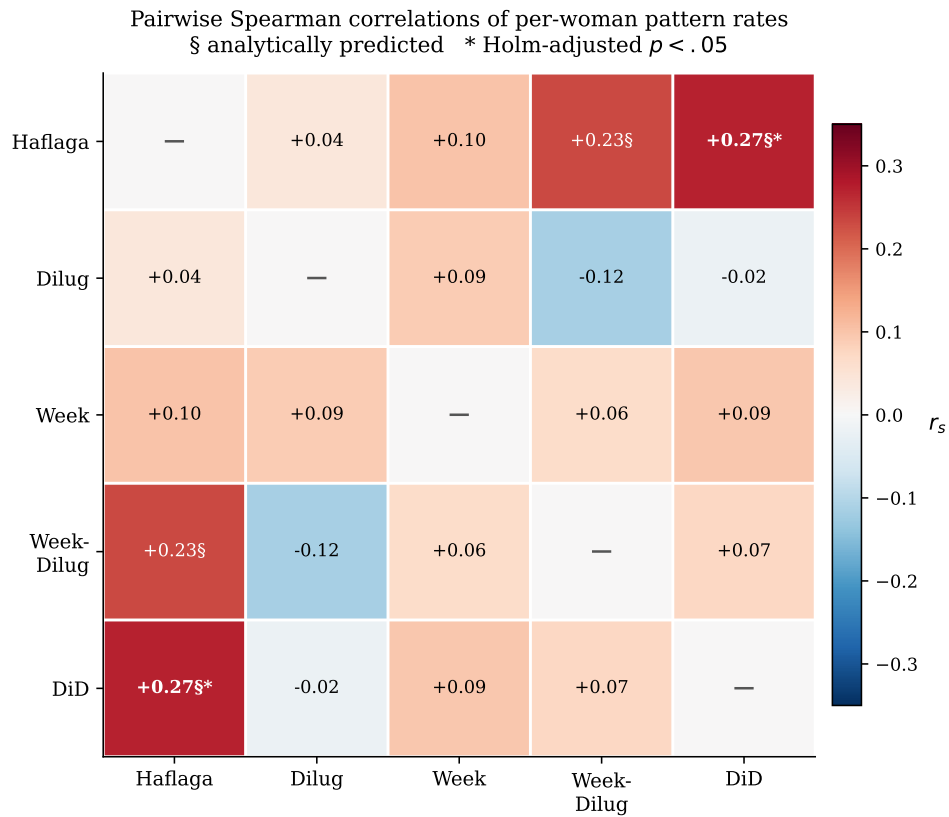


Figure S1: Spearman correlations between per-woman pattern rates. §: analytically expected positive pair; *: Holm-adjusted $p < .05$. All pairwise tests are exploratory.

S1.4 Robustness to follow-up length

Because both rates in a pair can depend on follow-up, the unrestricted permutation tests marginal rather than follow-up-conditional independence. Two checks, computed by the same script with the same seed, bound the consequence for the two structurally expected pairs. First, the partial Spearman correlation given n_i (correlation of the rank residuals after regressing each rank vector on the rank of n_i) is 0.254 for Haflaga $\times$ Dilug-in-Dilug (unadjusted 0.269) and 0.239 for Haflaga $\times$ Week-Dilug (unadjusted 0.231). Second, restricting the 50,000 permutations to exchange rates only among women in the same follow-up quintile (strata of 24, 8, 31, 27 and 28 women) preserves each rate's dependence on n_i while breaking the association between patterns; the resulting two-sided p -values are .0046 for Haflaga $\times$ Dilug-in-Dilug (unrestricted .0036) and .0097 for Haflaga $\times$ Week-Dilug (unrestricted .0113). The exploratory conclusions are therefore not an artefact of unequal follow-up. These checks are reported as robustness diagnostics for an exploratory analysis, not as an additional primary result.

S1.5 Interpretation

Three cautions apply, and a fourth follows from the external replication: the Haflaga $\times$ Dilug-in-Dilug association did not replicate in the Utah cohort ($r_s = -.09$; Section S4.6). The associations are between women, not within sequences, and say nothing about temporal ordering. They involve rare patterns (18 Week-Dilug and 27 Dilug-in-Dilug events) and rest on few women. And the surviving association is compatible with the proximity argument but is not evidence for it: attributing co-occurrence to a mechanism would require an explicit generative model for the sequences, which the article does not fit.

S2. Literature Search for Prior Randomisation-Based Tests

The main article states that we are not aware of any formal randomisation-based test of whether the frequency of vessel-type arithmetic patterns in empirical cycle data departs from a chance null. A hit was taken to contradict this only if it used empirical cycle data, defined rule-based arithmetic events (equal intervals, constant first or second differences, weekly anchors or

equivalent) and compared their observed frequency with a randomisation, permutation, bootstrap or Monte Carlo null. Descriptive halachic studies, forecasting models, Markov models and regularity statistics were not counted.

The search had two stages. First, an unstructured search of the Hebrew and English halachic, medical-halachic and biostatistical literatures carried out by the author while preparing the dissertation (2019–2022); the quantitative halachic works it located are cited in Section 2.3 of the main article and none tests pattern frequencies against a randomisation null. Second, a logged web search on 16 September 2026 of twelve queries in English and Hebrew combining the halachic terms (*vesset*, *veset*, *haflaga*, *dilug*, *onah*, *niddah*) with statistical terms (permutation, randomisation, Monte Carlo, probability, empirical data), together with citation-chasing queries for the author’s earlier articles and for re-analyses of the Marquette dataset. Every returned item was screened by title and snippet. No publication meeting the criterion was found in either stage. The verbatim queries, hit descriptions and limitations of the search (general web index rather than bibliographic databases; snippet-level screening; possible unindexed Hebrew material) are recorded in `revision_reports/literature_search_log.md` in the code repository. The main article therefore retains the wording “we are not aware of” rather than asserting absence.

S3. The Zero-Count Rule *Haflaga Chozer Chalila* Under the Three Nulls

Definition. — *Haflaga Chozer Chalila* (“recurring haflaga”) is established when a block of $m \geq 3$ consecutive intervals is repeated exactly, three times in succession, so that the event occupies a window of $3m \geq 9$ cycles. Thus $(25, 27, 26, 25, 27, 26, 25, 27, 26)$ is an event, $(25, 27, 26, 28)$ repeated three times is an event with $m = 4$, and $(25, 27, 25, 27, 25, 27)$ is not (block of length 2). It is the sixth and last rule computable from interval lengths alone; the other repeating rule of the halachic literature, *Dilug Chozer Chalila*, depends on calendar dates and is not computable from these data.

Counting convention. — As for the five primary rules, distinct maximal m -periodic segments of length at least $3m$ count once each. A segment whose block has a shorter period d dividing m is not counted at m : it is the same event at period d if $d \geq 3$, a *Haflaga* if $d = 1$, and not

an event if $d = 2$. The rule is implemented twice (a reference loop and a vectorised version) in `scripts/hcc_rule.py`, and the two implementations agree on the observed data and on 3,000 random sequences with planted repeating blocks. In the observed data 100 of the 118 women have the nine or more cycles needed for an event; the observed count is 0.

Null distributions. — The rule was counted under H_G , H_{iid} and H_W with $B = 50,000$ replicates each (seed 17), using the same generators as the main analysis (Table S3). Under the two global nulls an event occurs in fewer than 1 in 2,000 replicates; under H_W , which preserves each woman’s multiset of values, a periodic reordering of a woman with few distinct values and many cycles is occasionally produced, and at least one event occurs in 2.8% of replicates. The observed zero is the modal outcome under every null and its two-sided Monte Carlo p -value is 1. The replicate counts are stored in `results/hcc_null.npz`.

Effect on the joint statistic. — Because a near-degenerate component could in principle disturb the estimated covariance of the joint test, the check was made directly. The three null streams were regenerated with the seed and call order of the main analysis, so that the five primary counts reproduce the stored replicates exactly, and the rule was counted on the same permuted sequences, giving paired six-component replicate vectors (`scripts/hcc_joint.py`; `results/hcc_joint.npz`). Table S2 compares the five- and six-component Mahalanobis tests. The observed D_M is unchanged to two decimals under every null, because the observed deviation of the sixth component from its null mean is essentially zero. The p -values increase slightly: every replicate that contains an event lies far beyond the observed vector in the sixth coordinate, so such replicates are added to the exceedance count (16, 19 and 1,378 replicates under H_G , H_{iid} and H_W). The split-batch estimate agrees. Under the two global nulls the joint test still rejects at $p < .001$; under H_W the p -value moves from .126 to .150 and the test still does not reject. The condition numbers of the estimated 6×6 covariance matrices (1.4×10^5 , 1.2×10^5 and 2.0×10^3) reflect the near-zero variance of the sixth component but caused no numerical difficulty.

Table S2: Joint Mahalanobis test with five components (main article) and with *Haflaga Chozer Chalila* added as a sixth component, on the same $B = 50,000$ paired replicates per null.

Null	$D_M^{(5)}$	$p^{(5)}$	$D_M^{(6)}$	$p^{(6)}$	Split-batch $D_M^{(6)}, p^{(6)}$
H_G	5.08	.0001	5.08	.0005	5.07, .0005
H_{iid}	4.87	.0004	4.87	.0007	4.87, .0007
H_W	2.93	.126	2.94	.150	2.96, .143

Table S3: *Haflaga Chozer Chalila* under the three nulls ($B = 50,000$ each; observed count 0).

Null	Mean	SD	Max	$P(\text{count} \geq 1)$	p (two-sided)
H_G	0.0003	0.018	1	0.0003	1.00
H_{iid}	0.0004	0.020	1	0.0004	1.00
H_W	0.028	0.166	2	0.028	1.00

S4. External Replication in the Utah Creighton Model Cohort

S4.1 Source and preprocessing

The dataset is *Menstrual Cycles Length of Women in the USA and Canada, 1990–2013* (Stanford & Najmabadi, 2023; The Hive, University of Utah; <https://doi.org/10.7278/S50d-4gxs-s4hj>; licence CC BY-NC), which gives the start and end dates of 3,324 cycles from 581 women aged 18–40 with regular bleeding, not pregnant, not exclusively breastfeeding and without known subfertility, pooled from three Creighton Model FertilityCare cohorts: CMFS (1990–1996, 162 analysed women), TTP (2003–2006, 9) and CEIBA (2009–2013, 99); the cohorts are described by Najmabadi et al. (2020, *Paediatric and Perinatal Epidemiology* 34, 318–327). The deposit was downloaded by the author and is archived in the repository with its documentation (Table S4). It contains no duplicate rows; the recorded cycle length equals end – start + 1 for every cycle with a length; 2,557 of 2,743 within-woman successive pairs abut and 186 have a date gap, each coinciding with a jump in the cycle number (a cycle excluded in the source).

The preprocessing rules, fixed before any pattern was counted, are implemented in `external_replication/scripts/prepare_utah.py`: (1) cycle length L is the recorded length, $H = L + 1$; (2) the 180 conception cycles, all terminal and without a recorded

length, are excluded, and 7 cycles flagged “missing” but with a recorded length matching the dates are kept; (3) a woman’s record is split at every date gap (170 breaks among usable cycles) and gaps are never bridged; (4) each woman contributes her longest run of at least five consecutive cycles (earliest on ties; 7 women had more than one eligible run); (5) no cycle-length criterion is applied. Result: 270 women, 2,139 cycles, 5–15 cycles per woman ($M = 7.9$, SD 2.7, median 7); L 17–98 days ($M = 30.0$, SD 6.2, median 29; 12.0% of cycles ≥ 35 days); age 19–40 ($M = 27.2$, SD 4.4). Code, seed (17), $B = 50,000$ and the simulation sizes are those of the main analysis; the Week anchor set over the observed range is $\{22, 29, \dots, 99\}$.

Table S4: Files of the external replication and their SHA-256 checksums.

hive_download.zip (the deposit as downloaded)
ae0c63c3584c248bc10b141f49dbd9df34f550d56c8aff30dbb15471d455d71c
CrMcyclelength_share.csv (cycle table)
ad67a3eed731a038d6502a1eb42a9dcddf8f6c69a607a61b85f3798198043dac
Na jmabadi_README20230824.txt (documentation)
4bbb2c93efada3c84fe841bc5ef82820c30f8c1dce7e7ebc07d98558138a4678
sequences.csv (analysed sequences after preprocessing)
08dddb0c04c6a107b95eefc6ef081fabedfe2e22bd8c9832ce6e1af45a9ee37f

S4.2 Marginal and joint results

The observed counts are (41, 76, 35, 40, 41) for (Haflaga, Dilug, Week, Week-Dilug, Dilug-in-Dilug); 38, 66, 33, 35 and 39 women establish each pattern. Table S5 gives the marginal results under the three nulls and Figure S2 the null distributions; Table S6 gives the joint statistics next to the Marquette values, and Figure S3 compares the standardised deviations of the two cohorts.

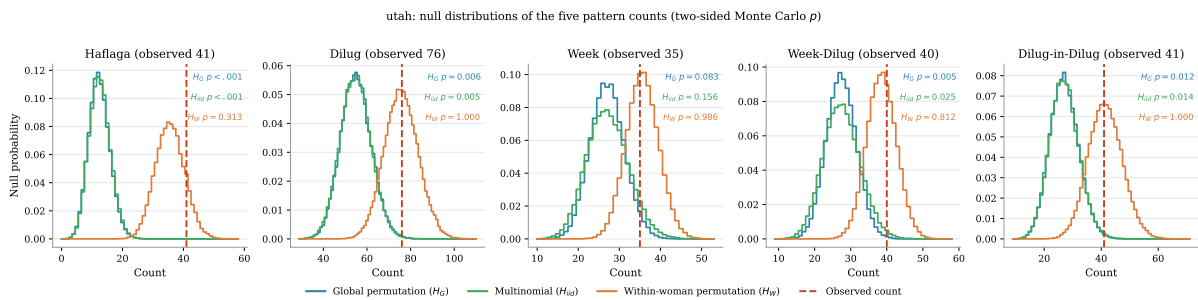


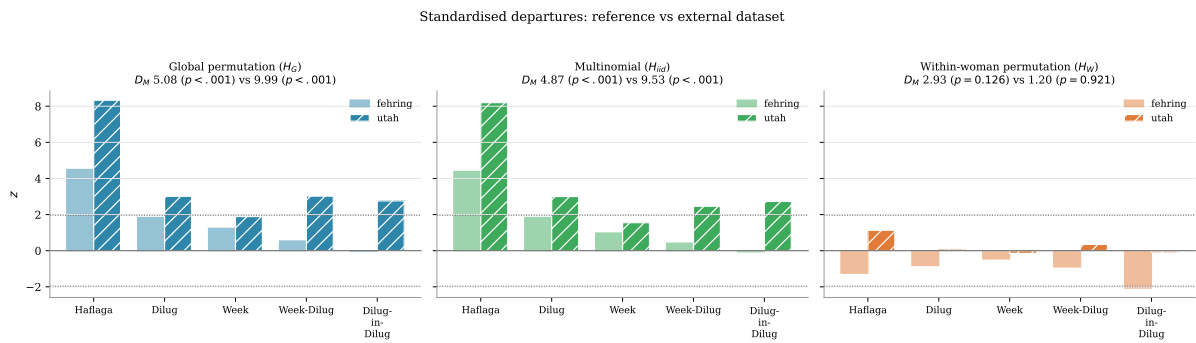
Figure S2: Utah cohort: null distributions of the five counts under H_G , H_{iid} and H_W with the observed counts marked; two-sided Monte Carlo p -values.

Table S5: Utah cohort: marginal results under the three nulls ($B = 50,000$; two-sided Monte Carlo p ; Holm over the five patterns within each null).

Null	Pattern	Obs	μ (σ)	z	p	Holm
H_G	Haflaga	41	12.42 (3.43)	+8.33	< .001	< .001
	Dilug	76	54.95 (6.99)	+3.01	.006	.021
	Week	35	27.09 (4.18)	+1.89	.083	.083
	Week-Dilug	40	27.40 (4.17)	+3.02	.005	.021
	Dilug-in-Dilug	41	27.08 (5.04)	+2.76	.012	.024
H_{iid}	Haflaga	41	12.56 (3.47)	+8.20	< .001	< .001
	Dilug	76	54.84 (7.05)	+3.00	.005	.021
	Week	35	27.17 (5.05)	+1.55	.156	.156
	Week-Dilug	40	27.52 (5.08)	+2.46	.025	.050
	Dilug-in-Dilug	41	27.09 (5.09)	+2.73	.014	.042
H_W	Haflaga	41	35.60 (4.78)	+1.13	.313	1.00
	Dilug	76	75.71 (7.75)	+0.04	1.00	1.00
	Week	35	35.61 (3.97)	-0.15	.986	1.00
	Week-Dilug	40	38.56 (4.10)	+0.35	.812	1.00
	Dilug-in-Dilug	41	41.48 (5.95)	-0.08	1.00	1.00

Table S6: Joint Mahalanobis statistics in the two cohorts ($B = 50,000$ per null). Three-pattern: exploratory Haflaga, Week-Dilug, Dilug-in-Dilug subspace.

Null	Marquette D_M (p)	Utah D_M (p)	Utah split-batch	Utah three-pattern
H_G	5.08 (< .001)	9.99 (< .001)	10.03 (< .001)	9.10 (< .001)
H_{iid}	4.87 (< .001)	9.53 (< .001)	9.52 (< .001)	8.81 (< .001)
H_W	2.93 (.126)	1.20 (.921)	1.20 (.922)	1.17 (.715)

**Figure S3:** Standardised deviations $z = (O - \mu)/\sigma$ under each null in the Marquette sample and the Utah cohort.

S4.3 Sensitivity analyses

Leaving out any single woman moves the H_W joint p -value within .861–.974 (Marquette .055–.224); no marginal p falls below .05 for any omission (Haflaga .248–.467). No woman has 20 or more cycles; the top follow-up quintile ($n_i \geq 11$, 56 women) holds 31.7% of cycles and 56% of Haflaga events. Table S7 gives the truncation and support-restriction analyses; in the latter every removed cycle sits at a sequence end, so no woman’s record is split, and the observed counts are unchanged. The restriction harmonises the support with the Marquette data and is not an inclusion criterion of either study.

Table S7: Utah cohort: sensitivity of the joint test to follow-up and support.

Analysis	Women	Cycles	Counts	$D_M(H_W)$	p
Primary	270	2,139	(41, 76, 35, 40, 41)	1.20	.921
First 12 cycles per woman	270	2,109	(40, 75, 34, 36, 41)	1.34	.878
Support $18 \leq L \leq 54$	258	2,050	(41, 76, 35, 40, 41)	1.15	.933

Under the support restriction the global nulls give $D_M = 9.27$ (H_G) and 8.85 (H_{iid}), both $p < .001$.

S4.4 Simulation validation

The size and power studies of the main article were re-run at full size on the Utah sequences (10,000 fresh datasets against 20,000 reference replicates for size; 400 datasets $\times$ 2,000 permutations per alternative for power). Empirical size is at or below nominal for every procedure (Table S8; Monte Carlo SE $\approx .002$ at .05), and the power of the H_W tests (Table S9) is close to that in the Marquette data (joint power .507, .995, 1.00 at $\rho = .25, .50, .75$ and .983, 1.00, 1.00 at $q = .10, .20, .30$).

Table S8: Utah cohort: empirical size at $\alpha = .05$ (10,000 datasets).

Null	Marginal (Haf, Dil, Wk, W-D, DiD)	Holm	Joint plug-in	Joint split
H_G	.039, .042, .030, .032, .046	.039	.051	.051
H_{iid}	.028, .039, .038, .036, .048	.038	.052	.046
H_W	.045, .042, .044, .036, .035	.034	.049	.051

Table S9: Utah cohort: power of the H_W tests at $\alpha = .05$ (400 datasets $\times$ 2,000 permutations).

Alternative	Haf	Dil	Wk	W-D	DiD	Holm-any	Joint
AR(1) $\rho = 0$	.030	.048	.035	.032	.035	.030	.045
$\rho = .25$	.135	.278	.110	.060	.132	.223	.443
$\rho = .50$	.682	.818	.358	.200	.465	.877	.993
$\rho = .75$	1.00	.992	.778	.595	.585	1.00	1.00
Persistence $q = 0$	.045	.025	.030	.058	.035	.028	.058
$q = .10$	.960	.180	.540	.360	.058	.938	.970
$q = .20$	1.00	.422	.942	.782	.070	1.00	1.00
$q = .30$	1.00	.718	.988	.908	.362	1.00	1.00

S4.5 Decomposition of the global standardised deviation

For each pattern the global z decomposes exactly as $z_G = (\mu_W - \mu_G)/\sigma_G + (O - \mu_W)/\sigma_G$: a heterogeneity gap between the within-woman and global null means, and a within-woman residual (Table S10). The gaps are of similar size in the two cohorts; the residuals are negative throughout in the Marquette data and near zero or positive in the Utah cohort, which is why the same heterogeneity produces a larger global displacement there.

Table S10: Decomposition $z_G = \text{gap} + \text{residual}$, with $\text{gap} = (\mu_W - \mu_G)/\sigma_G$ and residual $= (O - \mu_W)/\sigma_G$.

Pattern	Marquette			Utah		
	gap	residual	z_G	gap	residual	z_G
Haflaga	6.28	-1.79	4.50	6.76	+1.57	8.33
Dilug	2.76	-0.92	1.84	2.97	+0.04	3.01
Week	1.69	-0.45	1.24	2.04	-0.15	1.89
Week-Dilug	1.43	-0.90	0.54	2.68	+0.35	3.02
Dilug-in-Dilug	2.39	-2.42	-0.03	2.86	-0.10	2.76

S4.6 Exploratory pairwise associations

The rate-based Spearman analysis of Section S1 was repeated (Table S11). The Marquette association Haflaga $\times$ Dilug-in-Dilug did not replicate ($r_s = -.09$, $p = .14$; partial r_s given $n_i = -.14$; follow-up-stratified $p = .21$). The only non-null pair in the Utah cohort is Haflaga $\times$ Week ($r_s = .27$, $p = .0001$, Holm $p = .001$; binary version $\chi^2 = 19.9$, Holm $p = .001$); a Haflaga run at a Week anchor value necessarily produces a Week event, so this association is

partly structural, in the same way as $\text{Haflaga-30} \subseteq \text{Week-Dilug}$. The 38 women with a Haflaga event had a mean within-woman SD of H of 1.89 (SD across women 1.37) against 4.61 (4.47) for the 124 women with no event of any type, a difference expected by construction.

Table S11: Pairwise Spearman correlations between per-woman pattern rates in the two cohorts (two-sided permutation p , 50,000 permutations; Holm over the ten pairs).

Pair	Marquette			Utah		
	r_s	p	Holm	r_s	p	Holm
Haflaga $\times$ Week	+.10	.277	1.00	+.27	.0001	.001
Haflaga $\times$ Dilug-in-Dilug	+.27	.004	.038	−.09	.145	1.00
Week-Dilug $\times$ Dilug-in-Dilug	+.07	.480	1.00	+.09	.172	1.00
Haflaga $\times$ Week-Dilug	+.23	.011	.102	+.09	.178	1.00
Dilug $\times$ Dilug-in-Dilug	−.02	.819	1.00	+.07	.223	1.00
Haflaga $\times$ Dilug	+.04	.668	1.00	−.07	.243	1.00
Dilug $\times$ Week	+.09	.337	1.00	−.05	.368	1.00
Dilug $\times$ Week-Dilug	−.12	.206	1.00	+.01	.886	1.00
Week $\times$ Week-Dilug	+.06	.476	1.00	−.01	.911	1.00
Week $\times$ Dilug-in-Dilug	+.09	.326	1.00	−.00	.980	1.00

S5. Null Distribution Functions Under the Two Global Nulls (Marquette Data)

Figure S4 shows the empirical distribution functions of the five counts under H_G and H_{iid} in the Marquette data, referred to in Section 4.1 of the main article; the two global nulls agree closely (largest difference in two-sided p -values 0.11, for Week, whose null SD is larger under H_{iid}).

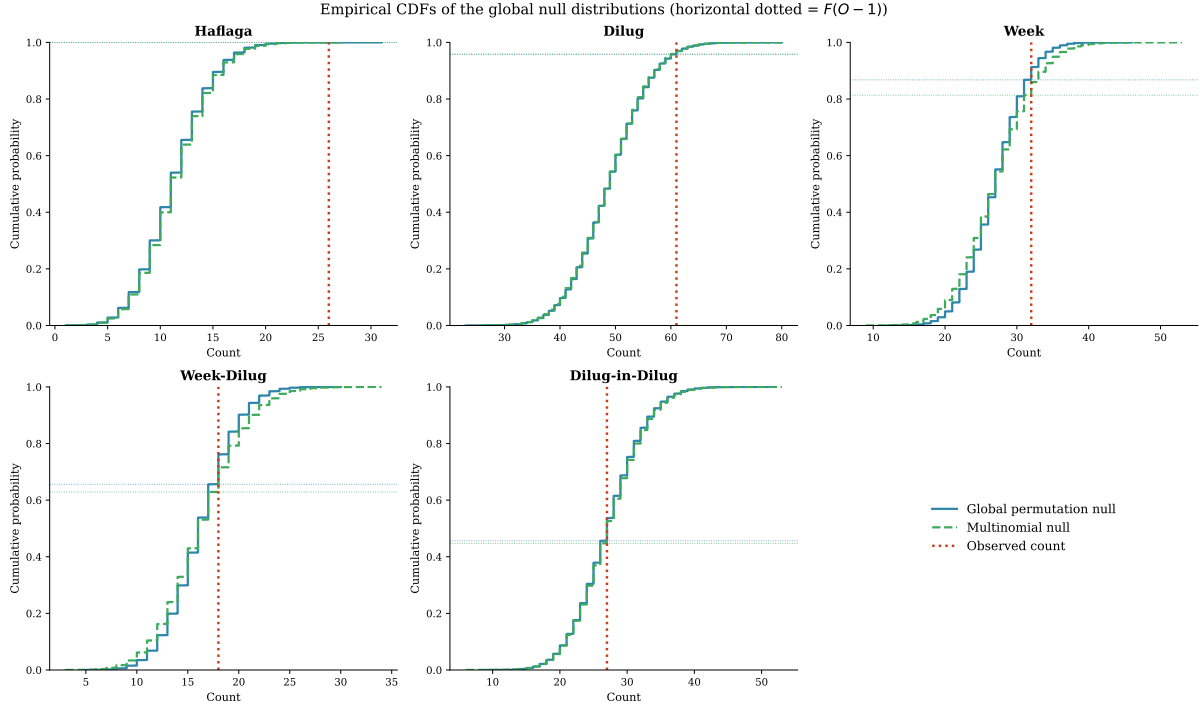


Figure S4: Empirical CDFs of the H_G (solid) and H_{iid} (dashed) null distributions in the Marquette data; dotted vertical line at the observed count.

S6. Regularity Strata and Per-Woman Concentration Under H_W

The aggregate H_W tests of the main article could in principle hide offsetting deviations in subgroups of women, or a within-woman excess confined to the few women who produce most events. Both possibilities were checked in each cohort using the per-woman replicate counts stored with the H_W replicates (`scripts/per_woman_hw.py`; results in `results/per_woman_hw.json` and `per_woman_hw_utah.json`). A woman's within-woman SD of intervals is a function of her multiset, which H_W preserves, so strata defined by it are fixed under the null and the stratified tests remain exact conditional Monte Carlo tests.

Regularity strata. — Women were split into terciles (and halves) of within-woman SD; observed and replicate counts were summed within each stratum and the marginal and joint tests recomputed (Table S12). No stratum in either cohort shows an excess of any pattern; every joint test is non-significant. The only marginal result surviving Holm correction within a stratum is a deficit, Dilug in the less regular half of the Marquette sample (13 observed against 25.1 expected, $z = -2.63$, Holm $p = .033$), which drives that stratum's joint p of .051; it is in the

same direction as the aggregate Dilug-in-Dilug deficit, is one of ten stratum-level analyses, and does not recur in the Utah cohort ($z = +0.72$). The least regular strata produce few events and therefore have little power against an excess.

Per-woman concentration. — For each woman the observed event total (all five patterns) was compared with her own H_W null distribution, and the distribution of events across women was compared with the null through the share of events held by the k most productive women *of each replicate*, the Gini coefficient of per-woman totals, and the number of women with no event (Table S13). In both cohorts the women who produce the most events are those whose multisets predict the most (rank correlation 0.62 and 0.70), no woman departs from her own null after multiplicity correction, and every concentration statistic lies at its null level. The concentration of events in a minority of regular women, which the global nulls register as excess, is therefore what H_W predicts woman by woman from the multisets alone.

Table S12: H_W tests within strata of within-woman SD of intervals (each stratum's counts are sums over its women; $B = 50,000$). Min Holm: smallest Holm-adjusted two-sided marginal p in the stratum.

Cohort	Stratum	n	Cycles	SD range	D_M	p	Min Holm
Marquette	most regular tercile	40	570	0.71–1.86	2.33	.37	.32
	middle tercile	39	511	1.86–2.81	1.57	.79	1.00
	least regular tercile	39	473	2.85–8.66	2.53	.26	.39
	below-median SD	59	796	0.71–2.19	2.69	.20	.13
	above-median SD	59	758	2.28–8.66	3.31	.051	.033 (Dilug deficit)
Utah	most regular tercile	90	741	0.52–1.77	2.00	.55	.80
	middle tercile	90	708	1.78–2.94	2.10	.49	1.00
	least regular tercile	90	690	2.98–26.9	0.73	.99	1.00
	below-median SD	135	1,110	0.52–2.16	1.40	.86	1.00
	above-median SD	135	1,029	2.16–26.9	1.60	.77	1.00

Table S13: Per-woman concentration of events under H_W . Overlap: women in both the top- k by observed events and the top- k by H_W expected events. Shares are of all events; the null value is the mean over replicates of the share held by the top- k women of that replicate; p two-sided Monte Carlo.

	Marquette (118 women)	Utah (270 women)
Spearman r_s , observed vs expected per-woman total	0.62	0.70
Overlap of top-5 / top-10 / top-20 sets	4 / 7 / 12	3 / 4 / 8
Women with per-woman two-sided $p < .05$ (expected)	1 (5.9)	1 (13.5)
Smallest Holm / Benjamini–Hochberg adjusted p	1.00 / 1.00	1.00 / 1.00
Per-woman z : mean (SD)	−0.23 (0.89)	0.02 (0.93)
Share of events, top 6 women: observed vs null (p)	.238 vs .208 (.12)	.107 vs .110 (.75)
Share of events, top 12 women: observed vs null (p)	.354 vs .336 (.40)	.185 vs .192 (.56)
Share of events, top 24 women: observed vs null (p)	.524 vs .526 (.98)	.313 vs .326 (.49)
Gini of per-woman totals: observed vs null (p)	.530 vs .515 (.59)	.595 vs .600 (.80)
Women with no event: observed vs null (p)	35 vs 31.1 (.41)	124 vs 125.3 (.90)